CD206
Business Programming

Integrated Library Management System

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Abstract

The Library Management System is developed to streamline and automate daily library operations, including member registration, book cataloguing, borrowing, and returning. The system provides a structured way to manage book collections and track issued and returned books in real time. By replacing traditional manual record-keeping with a digital platform, the system enhances operational efficiency, minimizes human errors, and improves accessibility of information. The overall goal is to provide a reliable and user-friendly solution that supports effective library management.

Objective

- To automate library book issuing and returning.
- To maintain student and book records in a database.
- To prevent duplicate IDs and incorrect borrowing.
- To calculate late return penalties automatically.
- To provide an easy-to-use interface.

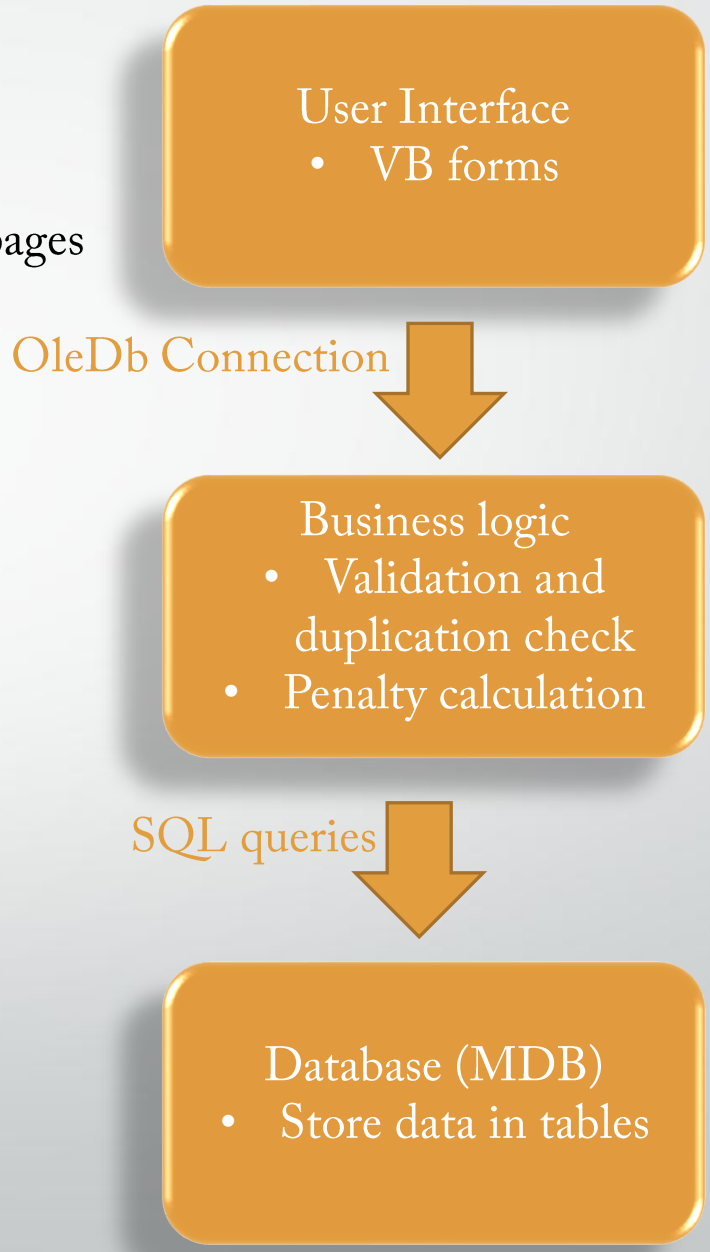
Architecture Diagram

User Interface Layer (Front-End)
•Developed using Visual Basic Forms
•Includes Login, Mainform, Students, Books, Issue and Return pages
•Users interact through buttons, textboxes, DataGridView

Business Logic Layer
•Handles input validation
•Prevents duplicate IDs; checks borrowing rules
•Controls navigation between forms

Database Layer (Back-End)
•MS Access database (Library1.mdb)
•Stores all records in tables
•Operations performed using SQL queries

Communication Between Layers
•VB uses OleDbConnection to connect to Access
•Queries are executed using OleDbCommand
•Results displayed using DataGridView



Module Explanation

Login And Registration

- Allow new users to register into the system
- Verify unique userID
- Username cannot be the same
- Login verifies username and password
- Successful login direct the main menu

Main Menu (Main Form)

- Acts as navigation hub of the system
- Buttons such as student, books, issue, return are avaliable
- Open corresponding page when button is clicked

Module Explanation

Student Management

- Add new student records
- Prevent duplicate Student ID
- Edit and delete existing records
- Display all students on DataGrindView

Books Management

- Add new books with unique Book ID
- Edit and delete book info
- Prevent duplicate book entries
- Display all books in DataGrindView

Module Explanation

Issue Books

- Allows student to borrow books
- Prevent duplication record
- Disallow different students not to borrow the same book at the same time
- Display list of all issued books in DataGrindView

Return Books

- Handles returning of borrowed books
- Calculates penalty based on due date
- Saves return record
- Deletes the issued record from issue table after return
- Display all books and student info on DataGrindView

Results

Library Management System Login



UCSI Library Management System Login

For new users

Username:

Password:

Register

For existing users

Username:

Password:

Login

Logout

Library Management System Main Page



Welcome to UCSI Library Management System

Students

Books

Issue

Return

Back

Student Page

Student

Student Details

Name:

Student ID:

Save | Edit | Delete | Reset

Student List

StudentID	Name
S0001	Chan Yi Xing
S0002	Lim Wei Jian
S0003	Nur Aina Sofea
S0004	Muhammad Hakim
S0005	Tan Jia Hui
S0006	Lee Zhi Ming
S0007	Priya Darshini
S0008	Amirul Haziq
S0009	Siti Nur Balqis
S0010	Chong Wei Lun
S0011	Ng Pei Wen

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Books Page

Books

Book Details

Book Title:

Year:

Book ID:

Author:

Save | Edit | Delete | Reset

Books List

BookID	Book Title	Author	Year
B0001	Java Programming	James Keller	2020
B0002	Introduction to Python	Sarah Thompson	2019
B0003	C++ for Beginners	Daniel Morgan	2021
B0004	Web Development Essentials	Laura Bennett	2018
B0005	Database Systems Basics	Michael Sanders	2020
B0006	Network Fundamentals	Olivia Carter	2017
B0007	Cloud Computing Explained	Richard Adams	2021
B0008	Artificial Intelligence Concepts	Emma Collins	2022
B0009	Machine Learning Guide	William Price	2020
B0010	Data Structures & Algorithms	Sophia Mitchell	2019
B0011	Mobile App Development	Brian Edwards	2018
B0012	Software Engineering Principles	Hannah Rogers	2021

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Issue Books Page

Issue Book

Issue Details

Student ID:

Book ID:

Issue Date: 10/12/2025

Name:

Book Title:

Save | Edit | Delete | Reset

Book Issued

	IssueID	StudentID	StudentName	BookID	BookTitle	IssueDate
▶	1	S0001	Chan Yi Xing	B0001	Java Programming	10/12/2025
*						

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Return Books Page

Return Book

Issue Details

	IssueID	StudentID	StudentName	BookID	Book Title	IssueDate	Penalty
▶	1	S0001	Chan Yi Xing	B0001	Java Programming	10/12/2025	0
*							

Return Details

	ReturnID	StudentID	StudentName	BookID	Book Title	IssueDate	ReturnDate	Penalty
*								

Issue Date: 10/12/2025 **Return Date:** 10/12/2025 Return

Conclusion

•The Library Management System successfully automates key library operations such as student management, book registration, issuing, and returning.

•The integration of Visual Basic forms with an MS Access database ensures efficient data storage and retrieval.

•Duplicate ID validation and borrowing rules improve data accuracy and prevent errors.

•The system provides a user-friendly interface, reduces manual workload, and increases overall efficiency in managing library records

Future Scope

- Add user roles (Admin, Staff, Student) with different permissions.
- Enable online access using cloud database.
- Add barcode or QR code scanning for faster issue/return.
- Generate printable receipts and monthly reports.
- Integrate email or SMS notification for due date reminders.

References

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